Probability and Statistics using R

Lab Session-1

Course Code : MAT445

Date: 16.07.2026 **Time : 3:15PM - 4:15 PM**

Questions

1. Write an R program to check whether a given integer is even or odd.
2. Write an R program to check whether a given integer is a prime number.
3. Write an R program to check whether a given integer is a palindrome.
4. Write an R program to find the factorial of a positive integer.
5. Write an R program to find the sum of digits of a given integer.
6. Write an R program to count the number of digits in a given integer.
7. Write an R program to find the Greatest Common Divisor (GCD) of two integers.
8. Write an R program to find the Least Common Multiple (LCM) of two integers.
9. Write an R program to check whether a given integer is an Armstrong number.
10. Write an R program to find the largest of two integers.
11. Write an R program to swap two integers without using a third variable.
12. Write an R program to check whether a given integer is a perfect number.
13. Write an R program to check whether a given integer is a perfect square.
14. Write an R program to check whether a number is positive, negative or zero.
15. Write an R program to generate the multiplication table of a given integer.
16. Write an R program to reverse the digits of a given integer.
17. Write an R program to find all factors of a given integer.
18. Write an R program to check whether two integers are coprime.
19. Write an R program to check whether a given integer is divisible by both 3 and 5.
20. Write an R program to compute a^b using a loop without using the exponent operator.

Solutions

1. Even or Odd

```
n <- as.integer(readline("Enter an integer: "))

if(n %% 2 == 0)
  cat("Even")
else
  cat("Odd")
```

2. Prime Number

```
n <- as.integer(readline("Enter an integer: "))

prime <- TRUE

if(n <= 1)
  prime <- FALSE

for(i in 2:(n-1)){
  if(n %% i == 0){
    prime <- FALSE
    break
  }
}

if(prime)
  cat("Prime")
else
  cat("Not Prime")
```

3. Palindrome

```
n <- as.integer(readline("Enter an integer: "))
temp <- n
rev <- 0

while(temp > 0){
  d <- temp %% 10
  rev <- rev * 10 + d
  temp <- temp %/% 10
}

if(n == rev)
  cat("Palindrome")
else
  cat("Not Palindrome")
```

4. Factorial

```r
n <- as.integer(readline("Enter an integer: "))

fact <- 1

for(i in 1:n)
  fact <- fact * i

cat("Factorial =", fact)
```

5. Sum of Digits

```r
n <- as.integer(readline("Enter an integer: "))

sum <- 0

while(n > 0){
  sum <- sum + n %% 10
  n <- n %/% 10
}

cat("Sum =", sum)
```

6. Count Digits

```r
n <- as.integer(readline("Enter an integer: "))

count <- 0

while(n > 0){
  count <- count + 1
  n <- n %/% 10
}

cat("Digits =", count)
```

7. GCD

```
a <- as.integer(readline("Enter first number: "))
b <- as.integer(readline("Enter second number: "))

while(b != 0){
  r <- a %% b
  a <- b
  b <- r
}

cat("GCD =", a)
```

8. LCM

```
a <- as.integer(readline("Enter first number: "))
b <- as.integer(readline("Enter second number: "))

x <- a
y <- b

while(y != 0){
  r <- x %% y
  x <- y
  y <- r
}

lcm <- (a * b) / x

cat("LCM =", lcm)
```

9. Armstrong Number

```
n <- as.integer(readline("Enter an integer: "))

temp <- n
sum <- 0

while(temp > 0){
  d <- temp %% 10
  sum <- sum + d^3
  temp <- temp %/% 10
}

if(sum == n)
  cat("Armstrong")
else
  cat("Not Armstrong")
```

10. Largest of Two Numbers

```
a <- as.integer(readline("Enter first number: "))
b <- as.integer(readline("Enter second number: "))

if(a > b)
  cat(a)
else if(b > a)
  cat(b)
else
  cat("Both are equal")
```

11. Swap Without Third Variable

```
a <- as.integer(readline("Enter first number: "))
b <- as.integer(readline("Enter second number: "))

a <- a + b
b <- a - b
a <- a - b

cat(a, b)
```

12. Perfect Number

```
n <- as.integer(readline("Enter an integer: "))

sum <- 0

for(i in 1:(n-1)){
  if(n %% i == 0)
    sum <- sum + i
}

if(sum == n)
  cat("Perfect Number")
else
  cat("Not Perfect Number")
```

13. Perfect Square

```
n <- as.integer(readline("Enter an integer: "))

r <- sqrt(n)

if(r == floor(r))
  cat("Perfect Square")
else
  cat("Not Perfect Square")
```

14. Positive, Negative or Zero

```
n <- as.integer(readline("Enter an integer: "))

if(n > 0)
  cat("Positive")
else if(n < 0)
  cat("Negative")
else
  cat("Zero")
```

15. Multiplication Table

```
n <- as.integer(readline("Enter an integer: "))

for(i in 1:10)
  cat(n, "x", i, "=", n*i, "\n")
```

16. Reverse Digits

```
n <- as.integer(readline("Enter an integer: "))

rev <- 0

while(n > 0){
  rev <- rev * 10 + n %% 10
  n <- n %/% 10
}

cat(rev)
```

17. Factors of a Number

```
n <- as.integer(readline("Enter an integer: "))

for(i in 1:n){
  if(n %% i == 0)
    cat(i, " ")
}
```

18. Coprime Numbers

```
a <- as.integer(readline("Enter first number: "))
b <- as.integer(readline("Enter second number: "))

x <- a
y <- b

while(y != 0){
  r <- x %% y
  x <- y
  y <- r
}

if(x == 1)
  cat("Coprime")
else
  cat("Not Coprime")
```

19. Divisible by 3 and 5

```
n <- as.integer(readline("Enter an integer: "))

if(n %% 3 == 0 && n %% 5 == 0)
  cat("Divisible by both 3 and 5")
else
  cat("Not Divisible")
```

20. Compute Power Using Loop

```
a <- as.integer(readline("Enter base: "))
b <- as.integer(readline("Enter exponent: "))

power <- 1

for(i in 1:b)
  power <- power * a

cat("Power =", power)
```